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ToxinPred 3.0: An improved method for predicting the toxicity of peptides

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ABSTRACT

Toxicity emerges as a prominent challenge in the design of therapeutic peptides, causing the failure of numerous peptides during clinical trials. In 2013, our group developed ToxinPred, a computational method that has been extensively adopted by the scientific community for predicting peptide toxicity. In this paper, we propose a refined variant of ToxinPred that showcases improved reliability and accuracy in predicting peptide toxicity. Initially, we utilized a similarity/alignment-based approach employing BLAST to predict toxic peptides, which yielded satisfactory accuracy; however, the method suffered from inadequate coverage. Subsequently, we employed a motif-based approach using MERCI software to uncover specific patterns or motifs that are exclusively observed in toxic peptides. The search for these motifs in peptides allowed us to predict toxic peptides with a high level of specificity with poor sensitivity. To overcome the coverage limitations, we developed alignment-free methods using machine/deep learning techniques to balance sensitivity and specificity of prediction. Deep learning model (ANN - LSTM with fixed sequence length) developed using one-hot encoding achieved a maximum AUROC of 0.93 with MCC of 0.71 on an independent dataset. Machine learning model (extra tree) developed using compositional features of peptides achieved a maximum AUROC of 0.95 with MCC of 0.78. We also developed large language models and achieved maximum AUC of 0.93 using ESM2-t33. Finally, we developed hybrid or ensemble methods combining two or more methods to enhance performance. Our specific hybrid method, which combines a motif-based approach with a machine learning-based model, achieved a maximum AUROC of 0.98 with MCC 0.81 on an independent dataset. In this study, all models were trained and tested on 80 % of data using five-fold cross-validation and evaluated on the remaining 20 % of data called independent dataset. The evaluation of all methods on an independent dataset revealed that the method proposed in this study exhibited better performance than existing methods. To cater to the needs of the scientific community, we have developed a standalone software, pip package and web-based server ToxinPred3 (https://github.com/raghavagps/toxinpred3 and https://webs.iiitd.edu.in/raghava/toxinpred3/).

1. Introduction

Peptides are short chains of amino acids, which are the building blocks of proteins. These molecules often serve as hormones, neurotransmitters, growth factors, ion channel ligands, or anti-infectives, among other vital functions. Their versatile biological roles make them ideal candidates for therapeutic applications [1,2]. In 1922, insulin, a peptide-based drug, received approval for treating diabetic patients, proving to be a vital lifeline for those with diabetes. Initially, producing sufficient human insulin posed a challenge, leading to the use of animal-derived insulins. However, this challenge was successfully addressed with the advancement of recombinant technology. Until the 1980s, only five peptide drugs had managed to enter the pharmaceutical market, reflecting the relatively slow pace of progress in this field during that period. In recent decades, there has been a tremendous effort to develop peptide-based therapeutics, resulting in the discovery and approval of hundreds of peptide-based drugs [3–5]. Peptide based therapeutics have number of advantages over traditional small

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molecules based drugs particularly in managing problem of drug resistance. These peptides exhibit low immunogenicity, high specificity, effectiveness, and overall safety [6]. As of now, over 170 peptides are currently undergoing active clinical development, and there are numerous additional therapeutic lead peptides are undergoing preclinical investigations [5,7]. Advances in areas such as structural biology, recombinant biologics, and modern synthetic and analytical techniques have significantly accelerated the process of designing peptide drugs [8].

Though experimental techniques for drug discovery are more accurate, they are time-consuming and extensive. It is important to use in silico or computer-aided drug discovery in order to improve the rate of drug discovery. In the field of peptide-based drugs, we use computer-aided peptide therapeutics to discover peptide based drug candidates. In the era of artificial intelligence, most of the tools heavily use machine learning or deep learning techniques for predicting potential peptide candidates for therapeutic use [9]. Over the years, a plethora of computational resources have been developed to support computer-aided peptide therapeutics, broadly falling into five major categories based on their functions (see Supplementary Table S1). The first category comprises resources developed to understand or predict the therapeutic properties of peptides, such as their half-life and structure [10–16]. The second category of resources developed to predict peptide candidates for drug delivery includes the prediction of cell-penetrating and tumor homing peptides [17–26]. In the last decade, a large number of resources have been specifically developed to understand the antimicrobial properties of peptides. These resources have been grouped into the third category, which includes resources on antiviral, antifungal, and antibacterial peptides [27–45]. The fourth category of resources is dedicated to peptide inhibitors, excluding antimicrobial peptides. Major resources in this category are associated with anticancer and antihypertensive peptides [46–57]. The fifth and final category contains resources associated with the harmful effects of peptides like toxic and hemolytic peptides [58–64].

One of the major objectives of this study is to predict the toxicity of peptides. Thus, we have summarized existing methods of toxicity. In the past, numerous methods have been developed for predicting the toxicity of peptides, including DeepTox Field [65], ProTox-II Field [66], and eToxPred [67]. In addition, methods have been developed to predict the type of toxicity, like BTXpred [68] and NTXpred Field [69], which were developed to classify bacterial toxins and neurotoxins, respectively. Following computational tools ClanTox [70], SpiderP [71], ToxClassifier [72], TOXIFY [72], and ToxDL [73] have been designed to remove toxins from peptides from animal sources. A machine-learning technique called NNTox was developed in 2019 [74] to identify protein toxicity based on several gene ontology annotations. Gupta et al. published a general technique called ToxinPred [75] in 2013 to predict the toxicity of peptides regardless of their source. The scientific community makes extensive use of this technique to predict peptide toxicity. Wei et al. developed deep learning-based methods such as ATSE [76] and ToxIBLT [77] to predict protein and peptide toxicity based on structural, evolutionary, and physicochemical sequence features. ToxinPred2 [78], ToxMVA [79], and CSM toxin [80] are recently developed tools to classify the toxic proteins only. Zhao et al. published deep learning based on different deep learning architectures, including CNN and gated recurrent units (CNN_BiGRU), to predict the toxicity of peptides and proteins [81].

Despite the numerous methods developed in the past for predicting peptide toxicity, there remains a persistent and challenging need to develop a method that can predict peptide toxicity with high precision. In this work, we have made a systematic attempt to develop highly accurate and reliable toxicity prediction. In order to achieve our objective, we used the largest possible dataset for developing our methods. In this study, our models have been trained and validated in a dataset that contains toxic 5518 and 5518 non-toxic peptides. This dataset is three times more than the dataset used in our previous method, ToxinPred. In addition to the largest possible dataset, we used state-of-the-art algorithms and an ensemble of methods to improve the accuracy of prediction. Firstly, we explore alignment-based approaches that include motif and similarity search. MERCI software [82] has been used for discovering toxicity-associated motifs; these motifs are used to identify toxic peptides. Similarly, a similarity-based approach has been used to predict toxic peptides. These alignment-based approaches provide high specificity with poor coverage. To complement the alignment-based method, we also developed alignment-free models. Secondly, we explore alignment-free approaches that include prediction models based on machine and deep learning. In addition, we implemented large language models like ESM2 [83], ProtBERT [84], and BioBERT [85] in alignment-free models. Finally, we combined more than one approach for exploiting their strength. Our hybrid method performs better than any individual method.

2. Material and methods

Creation and collection of the dataset.

2.1. Benchmark dataset

Previous studies have shown that a robust dataset is necessary to train a prediction model with good generalization properties [86]. A new peptide toxicity dataset is constructed in this study. We have collected short sequences of toxic proteins or peptides from different databases and studies that include Conoserver [87], DRAMP [88], CAMPR3 [28], dbAMP2.0 [89], YADAMP [90], DBAASP-v3 [91] and UniProt (released on 2 June 2021) [92]. In our study, we excluded proteins/peptides that exceeded 35 residues or contained non-natural amino acids. However, there is a chance that the SwissProt [93] database may still contain a number of toxic peptides obtained from different sources. Therefore, we removed any identical toxic proteins/peptides from the data. This led us to a set of 5518 unique toxic proteins/peptides. We also searched for toxic proteins/peptides in SwissProt using specific criteria such as the keywords “toxin” or “toxic” and limiting the search to reviewed entries. These proteins/peptides were regarded as positive datasets for toxic peptides. While retrieving well-annotated or experimentally validated toxic peptides is feasible, obtaining non-toxic ones is challenging. Therefore, we searched for protein/peptide sequences in UniProt to construct a negative or non-toxic dataset using the keywords ’NOT toxin NOT toxic AND reviewed: yes’. In our data extraction process, we specifically targeted protein or peptide sequences that were shorter than 35 amino acids in length and eliminated any sequences containing non-natural amino acids. By excluding non-natural amino acids, we focused solely on naturally occurring protein or peptide sequences, which are more representative of biological systems. To address redundancy in both the positive and negative datasets, we utilized Venny software [94] to ensure the removal of identical peptides. This resulted in a set of 4321 non-toxic peptides as negative datasets. It is important to ensure that the data used to train a toxicity detection model is balanced in terms of the number of positive and negative examples [95]. This is because a model that is trained on unbalanced data is more likely to overfit or underfit the data [96]. We randomly generated peptides from the SwissProt database to ensure a balanced representation of positive and negative datasets. Ultimately, we arrived at 5518 unique positive and negative datasets. The creation of datasets is depicted in Fig. 1.

2.2. BLAST search

Most of the existing methods have been developed or optimized for predicting similarity/homology between proteins. Basic Local Sequence Alignment Tool (BLAST), version 2.9.0, is used in this study to identify toxic peptides. This tool is often employed to discover and annotate nucleotide and protein sequences [97]. BLASTp-short is specifically

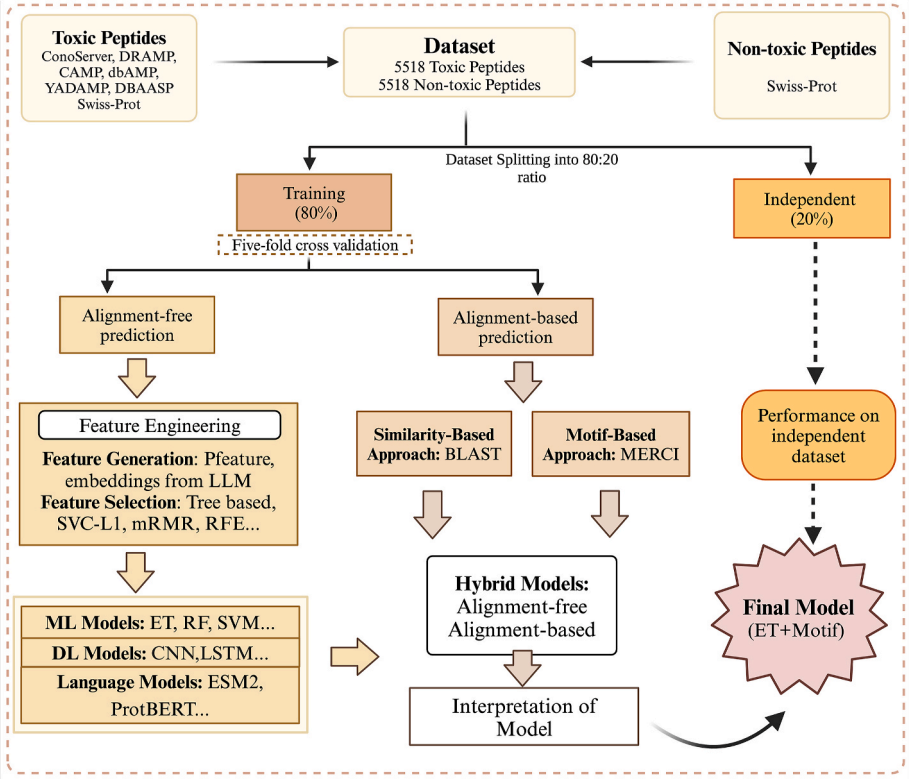


Fig. 1. Flowchart illustrating the sequential process for ToxinPred3.0.

optimized for predicting similarity between peptides, thus we used BLASTp-short in this study over existing methods. We have specifically used BLASTp-short, a method designed for identifying short proteins, i.e., peptides, based on the resemblance between the sequences. To accomplish our objective, our initial step involved constructing a repository consisting of both toxic and non-toxic peptides. Once the repository was established, we proceeded to hit the query sequences against this repository to categorize them as toxic or non-toxic. This procedure was performed for various e-value cut-offs ranging from 10^{-6} to 10^{2}. The repository was established by incorporating all the sequences present in the training dataset. To assess the performance of the training dataset, we took the first hit after the self-hit into account. On the other hand, when evaluating the independent dataset, the first hit was taken to establish whether a sequence is toxic or non-toxic. For example, if the query sequence hit a toxic peptide, the sequence was categorized as toxic, and similarly, for non-toxic peptides.

2.3. Motif analysis

In motif-based analysis, we identify small segments or patterns within peptides that contribute to their toxicity. Essentially, a motif represents a specific region within a peptide that is accountable for its toxic effects. Thus, motif analysis not only enhances prediction accuracy but also aids in identifying the precise motif or region responsible for toxicity. The Motif-EmeRging and with Classes-Identification (MERCI) tool was used to identify recurring patterns, or motifs, in the toxic proteins [82]. The MERCI tool uses a Perl script to search for motifs in a set of sequences using default parameters. This analysis provides information about the amino acid sequences that are associated with toxicity. The MERCI tool was used to analyze a set of 5518 toxic proteins. The analysis identified 117 motifs that were unique to toxic proteins. These motifs are used to assign weights, which were subsequently added to the prediction scores produced by the predictive model. The identification of motifs in toxic proteins is a valuable tool for understanding the molecular basis of toxicity. This information can be used to develop new drugs and therapies to treat toxic diseases. The MERCI tool has been applied extensively to detect motifs in various toxic proteins, including proteins involved in cancer, neurodegenerative diseases, and infectious diseases [82].

2.4. Deep learning (DL) based classifiers

In our study, we employed various deep learning architectures within the Keras framework to tackle the classification task. These architectures offer diverse ways of modeling sequential data. The initial model employs an Artificial Neural Network (ANN) with fixed-length LSTM layers, effectively capturing long-term dependencies by maintaining consistent sequence lengths. The second model, an ANN with LSTM layers and variable sequence lengths, adapts to real-world scenarios with varying sequence lengths. The third model combines an ANN with Bidirectional LSTM (Bi-LSTM) layers, enhancing temporal dependency comprehension by processing sequences in both directions. The fourth model, an RNN with LSTM layers, excels at handling sequential data through time step memory. Lastly, the fifth model integrates a Convolutional Neural Network (CNN) with Regularized Dense layers, extracting local patterns and improving generalization through regularization. Notably, the distinction lies in sequence length handling—fixed lengths simplify architecture but may lose information, while variable lengths maintain data integrity but demand more computational resources. After carefully examining the behavior of all, we have used the best performing, balanced model to be incorporated in the standalone to accurately predict the toxic and non-toxic peptides.

2.5. Feature generation

We use the Pfeature [98] standalone package to generate features of datasets. We considered all composition-based features and calculated a vector comprising of 9163 features for each sequence in the dataset. Supplementary Table S2 tabulates the information for each dataset training and independent dataset.

2.6. Feature selection and ranking

Previous studies have shown that not all features carry equal significance. Consequently, one of the primary obstacles lies in effectively identifying and selecting the most relevant features from a vast pool of features. With this in mind, our study has delved into various feature selection techniques, including support vector classifier (SVC) with L1-based feature selection technique (SVC-L1), Tree-based feature selection (extra tree), recursive feature elimination (RFE), SelectKBest (k-best), variance threshold (VTh), and minimum Redundancy Maximum Relevance (mRMR) to address this challenge comprehensively. By implementing these methods, we aim to select the most significant set of features from a vast pool of 9163 features that bear the utmost relevance to our investigation. The top-ranked features obtained from each method are shown in Supplementary Table S3. However, when it comes to creating various machine learning prediction models, their feature selection technique performance is nearly on par with AAC and DPC -based machine learning prediction models.

2.7. Machine learning (ML) based classifiers

A variety of machine-learning techniques have been employed to differentiate between toxic and non-toxic peptides. In order to construct classification models, the following machine learning algorithms were utilized: Extra Tree (ET), Random Forest (RF), Multi-layer Perceptron Classifier (MLPC), Logistic Regression (LR), XGBoost (XGB), Decision Tree (DT), and Support Vector Machine (SVM). These classifiers underwent optimization by exploring a range of hyperparameters, and the most favorable outcomes were selected for integration into the final models. The comprehensive workflow of ToxinPred3.0, including the implementation, is visually represented in the accompanying Fig. 1.

2.8. Embeddings from protein language models

Protein language models have been developed due to recent natural language processing (NLP) developments. These models use single amino acids and their doublets or triplets as tokens or words and produce fixed-size vectors, or embeddings, representing a specific peptide [99]. These protein embeddings have been used as inputs for various tasks, such as structure prediction, novel sequence creation, and protein classification. In our study, we have implemented three popular large language models (LLMs), namely ESM-2, ProtBERT, and BioBERT, to generate embeddings for peptide sequences. ProtBERT and BioBERT are extensions of the BERT model [100], which are pre-trained on a large dataset of protein sequences in a self-supervised fashion. On the other hand, ESM-2 (Evolutionary Scale Modeling) is a transformer protein language model originally designed for protein structure prediction and is trained over sequences in the UniRef protein sequence database. These embeddings were fed to ML classifier in order to develop a classification model.

2.9. Classification using protein language models

This study employed three state-of-the-art large language learning models, including ESM-2, ProtBERT, and BioBERT. Past studies have shown that fine-tuning these models on their own data for sequence classification can lead to high accuracy in a classification task [101, 102]. Pre-trained language models were implemented by initially fine-tuning them on our training dataset, and the fine-tuned language model was used to identify toxic peptides. The fine-tuned model was then evaluated on the independent dataset.

2.10. Cross-validation and performance metrics

The datasets used in this study were divided 80:20, with 80 % of data (i.e., 4414 toxic and 4414 non-toxic peptides) used for training and the remaining 20 % data (i.e., 1104 toxic and 1104 non-toxic peptides), were set aside as an independent for validation. Stratified k-fold cross-validation was used in this instance to assess the machine-learning models using 80 % of the training data. While stratified k-fold cross-validation shares similarities with k-fold cross-validation alone, it differentiates itself by incorporating a stratified sampling approach rather than a random selection. As a result, stratified k-fold cross-validation ensures that the distribution of samples across different classes or categories remains proportional and representative within each fold. The training data is split into five portions for internal validation, four of which are used for training and the fifth for testing. Five repeats of the identical procedure are carried out in order to test each of the five folds at least once. The performance of various machine learning models was assessed by employing well-established evaluation criteria, such as threshold-dependent and independent metrics. Sensitivity, specificity, accuracy, and the Matthews correlation coefficient (MCC) are under the category of threshold-dependent metrics, as their values are influenced by the classification threshold used. On the other hand, the area under the receiver operating characteristic curve (AUC) remains independent of the threshold and provides a comprehensive measure of a model's discriminative ability. These evaluation standards have undergone extensive review and annotation in prior investigations, establishing their reliability and importance in evaluating model performance.

$$Sensitivity = \frac{TP}{TP + FN}$$

$$Specificity = \frac{TN}{TN + FP}$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

where *FP*, *FN*, *TP*, and *TN* are false positive, false negative, true positive, and true negative, respectively.

2.11. Ensemble approach for classification

In this study, we have incorporated an ensemble/hybrid approach to improve the predictive capabilities of our final model. This hybrid approach involves a weighted scoring method, combining two distinct methods: (i) A motif-based approach utilizing MERCI and (ii) deep learning/machine learning based technique. To begin with, we employed the MERCI motif-based approach to classify the given peptide sequence. Each prediction received a weight of '+0.5′ for positive predictions (indicating toxic peptides), '-0.5′ for negative predictions (indicating non-toxic peptides), and '0′ for cases where no matches were found. This weight assignment provides a quantitative measure of confidence in each prediction. In the hybrid approach, the scores obtained from both the MERCI method and the machine learning or deep learning scores were combined to calculate an overall score. Considering various threshold values, the peptide sequence could be classified as toxic or non-toxic based on this overall score. This hybrid methodology has been widely employed in numerous research studies, demonstrating its effectiveness in improving prediction accuracy. By combining the strengths of motif-based and machine learning/deep learning techniques, our hybrid approaches (MERCI + ML and MERCI + DL) provide a comprehensive and robust framework for peptide classification, enabling more accurate toxicity predictions.

3. Results

In this study, we employed a total of 5518 toxic peptides as a positive dataset and 5518 non-toxic peptides as a negative dataset. All the

subsequent analyses and predictions are performed on toxic and non-toxic peptides. Our result section has been broadly divided into the following subsections: i) Analysis of toxic peptides, ii) Alignment-based methods, iii) Alignment-free models, iv) Large language models, v) Hybrid methods, vi) Benchmarking, and vii) Service to community. Following is description of these subsections.

3.1. Analysis of toxic peptides

It is important to understand the difference in the sequence of toxic and non-toxic peptides for developing any prediction model. In this study, two type of analysis has been performed called composition analysis and positional analysis.

3.1.1. Compositional analysis

The investigation involved the calculation of amino acid composition (AAC) for both toxic and non-toxic peptides. Our analysis revealed that the average AAC of amino acid residues such as cysteine, histidine, proline, and asparagine exhibited a higher prevalence in the toxic sequences. In contrast, valine, threonine, arginine, glutamine, methionine, leucine, isoleucine, phenylalanine, and alanine are higher in the non-toxic sequences. Furthermore, we compared the average AAC between the peptides and proteins of ToxinPred and ToxinPred3.0. Notably, we observed a remarkable abundance of cysteine and proline residues in the peptides derived from ToxinPred. In ToxinPred3.0, we find the same trend in composition as ToxinPred. The comparison of average AAC between ToxinPred and ToxinPred3.0 is depicted in Fig. 2 (A).

3.1.2. Positional analysis

During this analysis, our focus is on investigating the preference of specific amino acids at particular positions within the peptide sequence. To accomplish this, we construct a Two logo sample (TSL) for both toxic (positive) and non-toxic (negative) peptides, as depicted in Fig. 2 (B). The TSL provides valuable information regarding the proportional representation or relative abundance of amino acid residues and their importance within the sequence. It is crucial to remember that the initial four positions of the peptide correspond to the N-terminal residues, while the final or last four positions correspond to the C-terminal residues. The representation of relative abundance is based on the amino acid residue with the highest significance. Upon analyzing the toxic peptides, we observe a consistent preference for the amino acid residue 'cysteine' across all positions. This finding indicates that 'cysteine' is favored both in the N-terminal and C-terminal residues. Furthermore, at positions 1 and 3 in the C-terminus of toxic peptides, we find a higher prevalence of the amino acid residue 'glycine', suggesting its preference in N-terminus residues. In contrast, when examining non-toxic peptides, we discover that the amino acid residue 'methionine' predominantly occupies the first position. This observation implies that 'methionine' is the preferred residue at the beginning of non-toxic peptides. By examining these patterns of amino acid preferences, we gain valuable insights into the distinct characteristics and composition of toxic and non-toxic peptides. The implications of these findings extend our understanding of the connection between peptide structure and toxicity, paving the way for potential applications in peptide design and drug development.

3.2. Alignment-based methods

Alignment is one of the commonly used approaches in the field of bioinformatics for annotating protein/nucleotide sequences. In this study, we tried two alignment-based approaches for annotating toxic peptides: i) BLAST-based similarity search and ii) Motif-based approach.

3.2.1. BLAST-based similarity search

The algorithm BLAST is often used to discover the function of a peptide or a protein on the basis of similarity search. In this study, we used BLAST to identify toxic and non-toxic peptides based on the similarity between the two sequences. We have used BLASTp-short for the e-values 10^{-6} to 10^{2}. The results for BLAST are explained in detail in Table 1. It was seen that BLAST was only able to cover a part of the sequences and also had a significant error rate (I-hits). The maximum number of sequences BLAST was able to cover was 2097 out of 2208 sequences in the independent dataset with an error rate of 19.79 % at e-value 10^{2}. When we tried to minimize the error by decreasing the e-value, it was found that the sequences' coverage was insufficient. At e-value 10^{-6}, BLAST was able to cover only about 691 sequences out of the 2208 sequences with an error rate of 2.14 %.

3.2.2. Motif approach

In motif-based analysis, we identify small segments or motifs or patterns responsible for toxicity. In simpler terms, a motif is a region within a peptide responsible for its toxicity. In this paper, we used MERCI software to detect the specific motifs present exclusively in toxic and non-toxic peptides. Initially, we conducted motif searches in both the positive and negative training data. Our analysis revealed that motifs such as 'GCCS, HPAC, CCSN, CSHP, and CCSH' were solely present in toxic peptides. We extracted 117 motifs from the positive training data and 317 motifs from the negative training data. The minimum occurrence of these motifs in both the positive and negative training datasets was 35 (f = 35). A complete list of motifs is provided in Supplementary Table S4. Following the motif search, we proceeded to compare these motifs against the test dataset, consisting of both toxic and non-toxic peptides. The positive motifs successfully identified only 639 peptides out of a total of 1104 in the positive test dataset. Similarly, the negative motifs were able to recognize only 353 peptides out of the total 1104 in the negative test dataset.

3.3. Alignment-free methods

The performance of the similarity-based method depends on the similarity of query peptides with annotated peptides; it fails in the absence of similarity. In this study, we employed a wide range of alignment-free prediction models, mainly using artificial intelligence. In the case of alignment-free methods, we developed deep learning and machine learning methods.

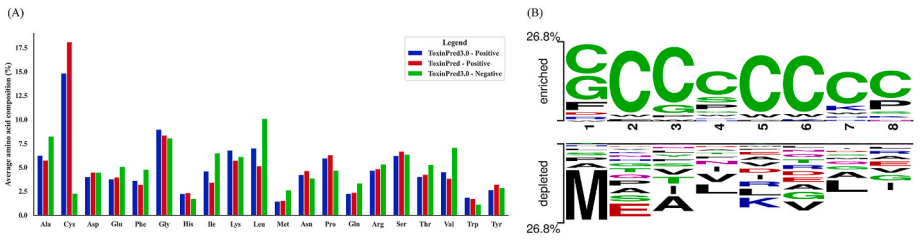


Fig. 2. (A) Comparison of average amino acid composition (AAC) between ToxinPred and ToxinPred3.0. (B) The preference of residues in toxic and non-toxic peptides at different positions in ToxinPred3.0 is illustrated in two logo samples. First, four positions serve the N-terminus of peptides, while the last four positions serve the C-terminus of peptides.

Table 1
BLAST similarity search results for training and independent dataset, where the blast database consists of sequences of the training dataset.

E-value	Training dataset					Independent dataset				
	Total hits	Toxic		Non-toxic		Total hits	Toxic		Non-toxic	
		C-hits (% of total hits)	I-hits (% of total hits)	C-hits (% of total hits)	I-hits (% of total hits)		C-hits (% of total hits)	I-hits (% of total hits)	C-hits (% of total hits)	I-hits (% of total hits)
10^{2}	8366	3785 (45.24 %)	550 (6.57 %)	2981 (35.63 %)	1050 (12.55 %)	2097	913 (43.54 %)	166 (7.92 %)	747 (35.62 %)	271 (12.92 %)
10^{1}	6008	3685 (61.33 %)	261 (4.34 %)	1745 (26.98 %)	317 (5.28 %)	1537	930 (60.51 %)	71 (4.62 %)	448 (29.15 %)	88 (5.73 %)
1	5241	3447 (65.77 %)	203 (3.87 %)	1414 (26.98 %)	177 (3.38 %)	1332	877 (65.84 %)	45 (3.38 %)	356 (26.73 %)	54 (4.05 %)
10^{-1}	4752	3172 (66.75 %)	171 (3.60 %)	1260 (26.52 %)	149 (3.14 %)	1228	823 (67.02 %)	30 (2.44 %)	330 (26.87 %)	45 (3.66 %)
10^{-2}	4269	2886 (67.34 %)	145 (3.40 %)	1105 (25.88 %)	133 (3.12 %)	1118	751 (67.17 %)	34 (3.04 %)	294 (26.30 %)	39 (3.49 %)
10^{-3}	3827	2577 (67.34 %)	135 (3.53 %)	995 (26 %)	120 (3.14 %)	991	663 (66.90 %)	31 (3.13 %)	262 (26.44 %)	35 (3.53 %)
10^{-4}	3394	2270 (66.88 %)	115 (3.39 %)	907 (26.72 %)	102 (3.01 %)	898	597 (66.48 %)	29 (3.23 %)	243 (27.06 %)	29 (3.23 %)
10^{-5}	2940	1934 (65.78 %)	105 (3.57 %)	811 (27.59 %)	90 (3.06 %)	787	513 (65.18 %)	30 (3.81 %)	221 (28.08 %)	23 (2.92 %)
10^{-6}	2569	1648 (64.15 %)	93 (3.62 %)	744 (28.96 %)	84 (3.27 %)	691	452 (65.41 %)	21 (3.04 %)	194 (28.08 %)	24 (3.47 %)

C-hits = correct hits, I-hits = incorrect hits.

3.3.1. Deep learning based models

In this study, we employed various deep learning models, including ANN with fixed and variable sequence length LSTM layers, ANN architecture with Bidirectional LSTM (Bi-LSTM) layers, Recurrent Neural Network (RNN) with LSTM layers and Convolutional Neural Network (CNN) with Regularized Dense layers that exercised L2 regularization. These models proved to be effective in retaining the relevant information from the training phase and adapting seamlessly to new sequences that contained diverse peptides and amino acids during the testing phase. The customized architecture of the model successfully comprehended the underlying nature of the data. The performance of all deep learning models used in the study is displayed in Table 2. ANN-LSTM with fixed sequence length achieved a maximum AUC 0.93, while ANN with Bi-LSTM layers achieved at least 0.90 AUC with balanced performance in training and testing. All the models achieved an impressive AUROC score, indicating strong predictive capability and high accuracy in distinguishing between positive and negative instances. However, it is worth highlighting that the machine learning model performed remarkably well in the classification of toxins, surpassing expectations.

3.3.2. Machine learning-based models

We create prediction models using a variety of machine-learning classifiers, including RF, LR, DT, ET, MLP, XGB, and SVM. The Pfeature compositional-based module was used to compute the features of the toxin and non-toxin peptides first. Using Pfeature, a total of 9163 features were calculated for each protein. We used various feature selection techniques to choose the most pertinent features, including SVC with L1 regularization, tree-based feature selection, K-best features based on statistical measures, RFE, mRMR, and VTh. Fig. 3 (A) visually represents how well the extra tree model performs, specifically in terms of the Area Under the Curve (AUC), when applied to an independent dataset. This assessment considers the use of different feature selection techniques. Among all the machine learning algorithms, tree-based classifier performed the best. It is evident that feature selection did not significantly improve performance after carefully analyzing how the ET model performed using different feature selection techniques and comparing them with results obtained without feature selection. As the majority of the features after feature selection belonged to the AAC and DPC category, we also trained our models using only the AAC and DPC compositional features and the performance of different models on the independent dataset shown in Fig. 3 (B). While using the features selected by the tree-based feature selection method, the time taken for generating the results was about 415.22s, while the time taken dropped to 50.58s when we used AAC and DPC features for the model. The model achieved similar performance without a significant loss in metrics while becoming time-efficient at the same time. Also in this case, ET managed to outperform other models with an AUROC of 0.94 and 0.95 in training and testing, respectively. Supplementary Table S5 elucidates a detailed comparison of the performance of several machine learning models, utilizing different feature selection methods, across both training and independent datasets.

3.3.3. Feature importance

Feature importance scores play a crucial role in understanding the significance of different features in a predictive model, thus enabling the construction of a more interpretable model. This score is calculated using the scikit-learn feature importance attribute, which is computed as

Table 2
Performance metrics of deep learning-based models utilizing one-hot encoding for ToxinPred3.0. Results highlight the effectiveness of the models in capturing intricate patterns and relationships.

DL models	Training dataset						Independent dataset					
	SENS	SPEC	PREC	ACC	MCC	AUC	SENS	SPEC	PREC	ACC	MCC	AUC
ANN with Bi-LSTM	0.89	0.89	0.89	0.85	0.78	0.98	0.86	0.86	0.86	0.86	0.71	0.93
RNN with LSTM layers	0.87	0.88	0.88	0.84	0.77	0.95	0.86	0.85	0.86	0.86	0.72	0.92
CNN with Relu-Dense layers	0.88	0.88	0.88	0.84	0.76	0.95	0.85	0.85	0.84	0.84	0.67	0.92
ANN - LSTM- Variable seq length	0.87	0.87	0.88	0.84	0.75	0.95	0.87	0.94	0.81	0.83	0.66	0.91
ANN - LSTM- Fixed seq length	0.90	0.91	0.90	0.85	0.80	0.91	0.83	0.84	0.83	0.83	0.67	0.90

SENS = sensitivity; SPEC = specificity; ACC = accuracy; AUC = area under receiver operating characteristic; MCC = Matthews correlation coefficient.

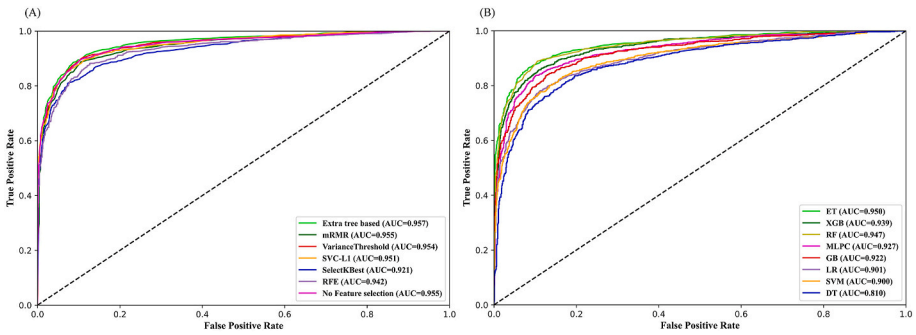


Fig. 3. Receiver Operating Characteristic (ROC) Curve comparing the performance of different ML models. (A) Performance of the Extra Tree Model with varied feature selection techniques on an independent dataset. (B) Performance of machine learning-based models developed solely using AAC and DPC composition features, measured by Area Under the Curve (AUC).

the mean and standard deviation of accumulation decrease in impurity, such as Gini impurity or entropy, caused by each feature in the trees within the model. In our study, which employs the Extra Trees algorithm, these scores elucidate how each feature contributes to the model's ability to discriminate between toxin and non-toxin peptides, and how they collectively enhance the model's performance. By plotting the accuracy of each feature against its importance score in the model, we aim to illustrate not just how well each feature can discriminate between toxin and non-toxin on its own, but also how it enhances the model's performance. For instance, the feature AAC_C prominently stands out, boasting an accuracy of approximately 80.34 % and a significance score of 0.077. This implies that AAC_C is the most critical feature for predictions in this dataset. Similarly, the DPC1_CC feature exhibits an accuracy of 74.64 % with a feature importance score of 0.044, making it the second most influential attribute. Fig. 4 shows the top 10 features which are contributing most to the model predictability.

3.4. Large language models

In last decade use of language models in bioinformatics have increased significantly. Number of protein and DNA language models have been develop to predict function of biological macromolecules. In this manuscript, we utilized protein language models ESM-2, ProtBERT, and BioBERT to predict toxicity of peptides. As these models are not tuned for classification of toxicity of peptides, thus we optimize hyperparameters of these models on our training dataset contain toxic and non-toxic peptides. The performance of tuned protein language models has been evaluated on an independent dataset (see Fig. 5A). Among protein language models, ESM2-t33 demonstrated superior performance, achieving an AUC of 0.93. In addition, we used these protein language models for generating embeddings which are used for developing machine learning models. We generated 1280, 320, 1024, and 767 embeddings from ESM2-t33, ESM2-t6, ProtBERT, and BioBERT, respectively for each peptide. In the case of ProtBERT, we also used a combination of a finetuned ProtBERT model with BiLSTM for extracting quality embedding. These embeddings are used as features for developing ML models for predicting the toxicity of peptides. Fig. 5B illustrates the performance of the top-performing machine learning model on an independent dataset, utilizing embeddings derived from various language model architectures. In this case, the ET model developed on ProtBERT + BiLSTM embedding achieved a maximum AUC of 0.957, better than other protein language models. Detailed performance metrics across training and independent datasets for each model are presented in Supplementary Table S6. It's noteworthy that the Extra Trees model, trained on features generated by Pfeature, demonstrates superior performance in terms of runtime efficiency compared to protein language models, with only a marginal difference in overall performance (see Fig. 5B).

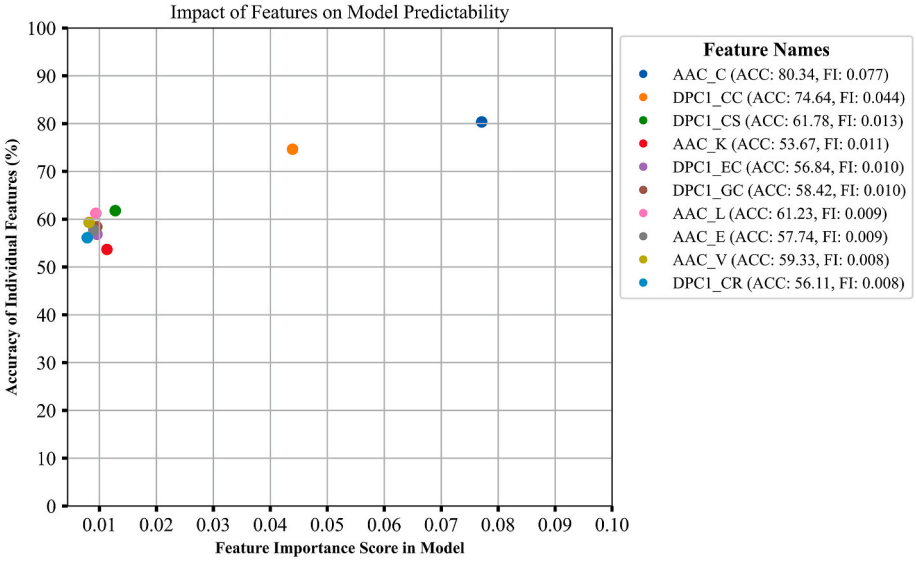


Fig. 4. Top 10 Features - Impact on Extra Trees Model Predictability. The figure illustrates the influence of key features on the Extra Trees Model, showcasing their significance based on both accuracy and importance scores (calculated using the feature_importance_ attribute of the model).
* ACC = Accuracy in %, FI = Feature Importance.

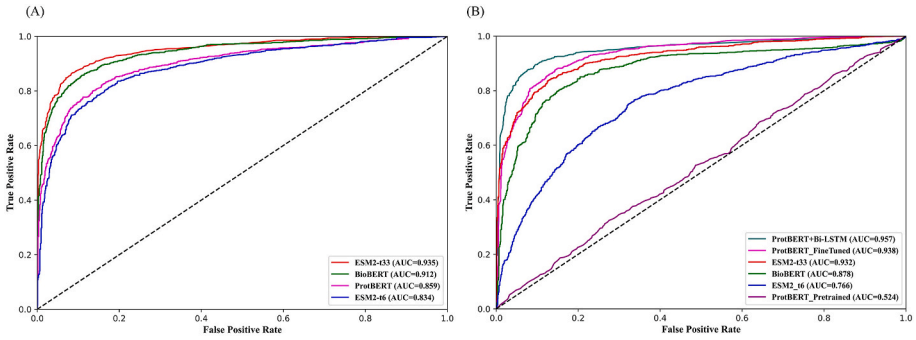


Fig. 5. Receiver operating characteristic (ROC) curve for large language models on independent dataset. (A) Performance of protein language model upon fine-tuning with our training data. (B) Comparative performance of Extra Tree model using features/embeddings of large language models.

3.5. Hybrid models

To enhance the predictive capability of our model, we employed a hybrid approach in this study. This approach utilized a weighted scoring method that combined two distinct techniques: i) Motif-based hybrid models, combining MERCI with deep learning and machine learning-based techniques, and ii) BLAST-based hybrid models, combining BLAST with deep learning and machine learning-based techniques.

3.5.1. Motif-based hybrid models

We have utilized an ensemble methodology to integrate the motif-based MERCI software with advanced deep learning and machine learning models to achieve precise prediction of protein toxicity. Initially, the peptide sequence provided was classified using MERCI. Positive predictions corresponding to toxic peptides were assigned a weight of +0.5, negative predictions for non-toxic peptides were assigned a weight of −0.5, and no hits were assigned a weight of 0. This innovative strategy leveraged the strengths of both approaches, resulting in a significant improvement in the AUC. Among all DL techniques we used, the ANN-LSTM - Fixed sequence length demonstrated the best performance on the independent dataset, achieving an AUC of 0.97 while in machine learning models, the ET (Extra Trees) model demonstrated exceptional performance on our independent dataset, achieving an AUC of 0.98 which was developed on composition-based features, specifically AAC and DPC (see Table 3). Notably, after combining the results of DL and MERCI, the performance comes very close to the ML and MERCI hybrid models.

Table 3
The performance of the motif-based hybrid approach on the independent dataset, when combined with different deep leaning and machine learning-based models.

Models		SENS	SPEC	PREC	ACC	MCC	AUC
DL	ANN - LSTM- Fixed seq length	0.92	0.91	0.90	0.92	0.81	0.97
	ANN - LSTM- Variable seq length	0.87	0.95	0.94	0.91	0.90	0.96
	ANN with Bi-LSTM layers	0.86	0.95	0.94	0.90	0.81	0.96
	CNN with Relu-Dense Layers	0.85	0.95	0.95	0.90	0.80	0.96
	RNN with LSTM layers	0.86	0.96	0.95	0.89	0.81	0.95
ML	ET	0.92	0.93	0.93	0.93	0.86	0.98
	RF	0.90	0.94	0.93	0.92	0.84	0.98
	XGB	0.89	0.94	0.94	0.91	0.84	0.98
	GB	0.89	0.92	0.92	0.91	0.82	0.97
	SVM	0.85	0.93	0.93	0.89	0.79	0.96
	MLPC	0.86	0.94	0.93	0.90	0.81	0.96
	LR	0.83	0.94	0.93	0.88	0.78	0.96
	DT	0.82	0.80	0.80	0.81	0.63	0.87

SENS = sensitivity; SPEC = specificity; ACC = accuracy; AUC = area under receiver operating characteristic; MCC = Matthews correlation coefficient.

We hereby understand that hybrid models perform really well in peptide sequence classification and can be used further in various vaccine or antibiotic generation processes. This research showcases the effectiveness of deep learning architecture and its future usage in bioinformatics.

3.5.2. BLAST-based hybrid models

To develop a robust prediction methodology, we integrated the similarity search method BLAST with separate machine learning (ML) and deep learning (DL) approaches. Initially, the BLAST search was conducted on a query sequence, and if a significant hit was found, the query sequence was classified as toxic or non-toxic based on the BLAST result. However, if the BLAST search did not yield any hits, an alternative approach, either ML or DL, was employed. We compared the performance of the ML-based approach to the DL-based approach combined with BLAST; we observed a decrease in performance for the ML model, which decreased from AUC 0.95 to 0.938 on the independent dataset. Conversely, the DL model demonstrated a slight improvement in performance, with the AUC increasing from 0.90 to 0.919. We further examined the performance of a combined approach involving ML or DL, the MERCI method, and BLAST. Unfortunately, combining BLAST with ML MERCI hybrid and DL MERCI hybrid did not improve the performance. It is worth noting that only the MERCI hybrid models achieved the best performance in our evaluation.

3.5.3. Final model

We developed a diverse array of models in this study, encompassing both alignment-free and alignment-based methods. Additionally, we designed hybrid methods that combine multiple approaches. Our two hybrid approaches achieved best performance: the first hybrid approach combined MERCI with ML techniques, while the second hybrid approach integrated MERCI with Deep Learning techniques. These two hybrid approaches were carefully designed and implemented to exploit the complementary strengths of both the involved methods. In addition, various large language models we developed, ESM2-t33, achieved maximum performance AUC 0.93, while in the case of embeddings extracted by these large language models, ProtBERT + Bi-LSTM embeddings performed best by achieving AUC 0.957 with the Extra Tree model. A comparison between the performance of the best ML, DL, LLMs, and hybrid models is shown in Fig. 6. Finally, we select extra tree + MERCI based ML hybrid and ANN-LSTM (fixed sequence length) + MERCI based DL hybrid as they demonstrated superior performance in terms of runtime efficiency compared to other models. We implement these hybrid models in our prediction software and web services for predicting and designing toxic peptides.

3.6. Benchmarking

It is important to benchmark a newly developed with existing methods to understand the importance of a newly developed method. In

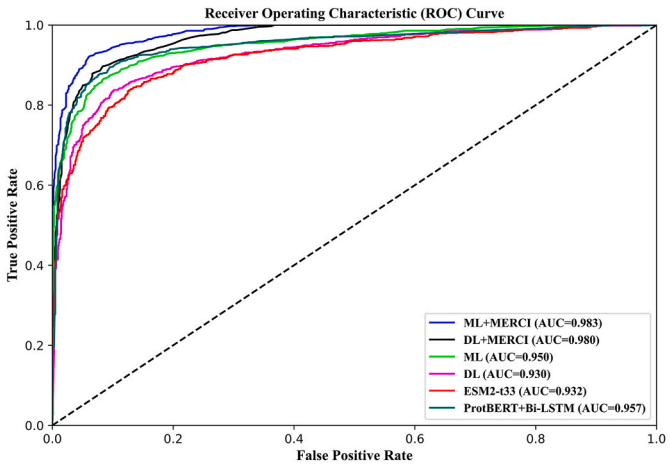


Fig. 6. ROC curves of different machine learning, deep learning, large language models, and hybrid models.

this study, we perform two types of benchmarking; i) Comparison of performance and ii) Ablation Study.

3.6.1. Comparison of performance

To support the creation of a new technique, it is crucial to evaluate how well the suggested method performs in comparison to already available methods. In Table 4, we compare the effectiveness of the suggested approach, ToxinPred3.0, with other currently used methods as stated in the literature. The data substantiates that ToxinPred3.0 surpasses alternative existing methods in performance.

We evaluated the effectiveness of current tools on the independent dataset used in our study to provide a comparison with other existing approaches. For the independent dataset utilized in ToxPred3.0, ToxinPred, and ToxIBTL obtained AUC values of 0.85 and 0.80, respectively, as stated in Table 4. The long protein toxicity prediction algorithms (ToxMVA, ToxinPred2, ToxClassifer, TOXIFY, and ToxDL) cannot be used because the dataset used in ToxinPred3.0 contains only peptide/short proteins sequences. Moreover, we were unable to predict the toxicity of proteins in the independent dataset of ToxinPred3 using ATSE and CNN_BiGRU due to the limitations of their services (ATSE - improper web service functioning, and CNN_BiGRU- standalone is not functional). The significance of the recently suggested strategy in the area of toxicity prediction is shown by this comparison in Table 4.

3.6.2. Ablation study

In the ablation study, we benchmarked all major models developed in this study, as summarized in Table 5. Firstly, we compared the performance of all models in terms of AUC, and the Extra Tree-based model demonstrated the highest AUC of 0.95. DL and protein language models achieved a maximum AUC of 0.93. Secondly, we assessed the time taken by these models for training and evaluation. The evaluation was conducted using the 'time' library from Scikit-learn on a machine equipped with an Intel(R) Xeon(R) Gold 6338 CPU @ 2.00 GHz, 100582 MB of RAM, and 16 cores. Decision tree-based models exhibited the shortest training and testing times compared to other models. In contrast, the BioBERT model took the longest time for both training and evaluation. As shown in Table 5, most protein language models required a considerable amount of time for training and testing. Additionally, we computed the size of all models and observed that decision trees had the smallest size, while esm2_t33_650 M_UR50D had the largest size (see Supplementary Table S7). Despite the decision tree's minimal training time and model size, its performance was relatively lower, with an AUC of 0.81. In this study, we selected the Extra Tree-based model, which exhibited the highest performance with a moderate training/evaluation time and model size.

Table 4
Comparison of Performance of existing methods on independent dataset of ToxinPred3.0

Methods	SENS	SPEC	PREC	ACC	MCC	AUC
ToxinPred	0.65	0.98	0.89	0.82	0.67	0.85
ToxIBTL	0.70	0.89	0.87	0.87	0.61	0.80
ToxinPred3.0	0.92	0.93	0.93	0.93	0.86	0.98

*SENS = sensitivity; SPEC = specificity; PREC = Precision; ACC = accuracy; AUC = area under receiver operating characteristic; MCC = Matthews correlation coefficient.

3.7. Service to the community

One of the major objectives of this study is to provide peptide toxicity prediction service to scientific community. Thus we developed a web server and standalone software package called ToxinPred3.0, which is available for scientific community from URL https://webs.iiitd.edu.in/raghava/toxinpred3/. The web server contain following major modules; a) Prediction, b) Protein scanning, c) motif scan and d) BLAST search. The "Prediction module" facilitates users to predict toxicity of their peptides, it allows user to submit multiple peptide sequences in FASTA format. The "Protein scanning module" allows one to identify or scan toxic regions in a protein. The "motif scan module" search toxic motifs or regions in peptide sequences submitted by a user. The utility of the 'BLAST search module' lies in its capacity to conduct similarity-based searches via BLAST against an extensive toxins and non-toxins database, enabling users to ascertain sequence similarities. In addition to web server, we also developed satandalone package which can be downloaded from our web server. A standalone software and pip package developed for large-scale toxin prediction, providing a versatile solution for comprehensive analysis.

4. Discussion

Peptide-based drug development requires reliable protein toxicity prediction to save time and costs [75]. This study aimed to create an accurate in-silico model for anticipating therapeutic peptide toxicity. The dataset included 5518 toxic peptides from databases and an equal number of non-toxic peptides from SwissProt. Various methods, like sequence alignment, motif recognition, and machine learning, were used for thorough analysis.

Our preliminary analysis of toxic peptides revealed a significant prevalence of Cys residues, consistently favored across most positions within the toxic peptide sequences. This observation was derived from the calculation of amino acid composition percentages. Furthermore, the amino acid composition of Pro, Asp, and His exhibited notable enrichment in toxic peptides in contrast to non-toxic counterparts. In the context of peptide analysis, dipeptide composition, which investigates the pairwise combinations of amino acids, has demonstrated superior predictive capability compared to composition-based models in prior research.

Utilizing the Pfeature tool, a comprehensive set of 9163 features was computed for each peptide sequence. The evaluation of our high-performing model encompassed three feature sets: the complete feature pool, AAC and DPC-based compositions, and a subset derived from extensive feature selection. Various selection techniques were employed to identify relevant features while eliminating irrelevant or redundant ones. Despite these efforts, the model's performance based on the selected features didn't exhibit significant enhancement. Remarkably, composition-based features consistently emerged as crucial discriminators between toxic and non-toxic peptides.

Our study aimed to enhance peptide toxicity prediction by integrating AAC and DPC information. Binary profiles were created, forming the basis for developing machine learning (ML) models. Notably, models combining AAC with DPC features outperformed those relying solely on

Table 5
Ablation study showing comparative timing in training and evaluation and size of model each model.

Model			Performance (AUC)	Training time(s)	Evaluation time (s)	Model size (MB)
ML	Extra Tree		0.950	240.2	50.8	52.7
	Random Forest		0.947	1725.4	113.6	8.5
	XGBoost		0.939	719.8	177.0	0.23
	Multi-layer Perceptron Classifier		0.927	739.0	143.6	14.6
	Gradient boosting		0.922	757.0	143.0	1.0
	Logistic Regression		0.901	905.6	183.1	0.96
	Support Vector Machine		0.900	943.0	118.0	0.14
	Decision Tree		0.810	18.52	0.12	0.11
DL	ANN-LSTM- Fixed seq length		0.932	720.0	25.0	1.48
	RNN with LSTM layers		0.920	515.5	4.6	0.82
	CNN with Relu-Dense layers		0.922	520.0	25.0	2.37
	ANN - LSTM- Variable seq length		0.913	98.04	1.73	21.82
	ANN with Bi-LSTM		0.900	201.43	18.6	0.63
LLM	ESM	esm2_t33_650 M_UR50D	0.930	2925.1	725.0	2480.0
		esm2_t6_8 M_UR50D	0.823	1809.0	567.4	35.0
	ProtBERT	Rostlab_prot_bert	0.861	1747.0	389.7	438.0
	BioBERT	biobert-base-cased-v1.1	0.912	3573.8	876.4	431.0
LLM	ESM	esm2_t33_650 M_UR50D	0.933	549.1	138.9	2560.8
Embeddings		esm2_t6_8 M_UR50D	0.850	269.0	60.8	100.1
	ProtBERT	Rostlab_prot_bert	0.957	925.4	365.1	1602.8
	BioBERT	biobert-base-cased-v1.1	0.880	518.3	120.34	450.1

basic amino acid composition, highlighting the importance of considering both composition and amino acid order. Additionally, we explored deep learning (DL) models using straightforward binary profiles, yet ML consistently outperformed DL in our evaluation.

Additionally, we integrated BLAST, a widely used tool for peptide sequence annotation. BLAST compares query peptides with an extensive database of known functions, assigning a function if a significant similarity is found. Employing BLASTp-short, we evaluated peptide toxicity. The algorithm covered more sequences at an e-value of 10^2, but with a 20.84 % error rate on an independent dataset. Conversely, at an e-value of 10^{-6}, coverage was lower with a 6.5 % error rate. BLAST alone couldn't accurately differentiate toxic and non-toxic peptides. Combining BLAST scores with Deep Learning and Machine Learning methods didn't yield improved accuracy due to BLAST's incorrect identifications (~2.14 %). Consequently, we excluded BLAST from the final hybrid models.

In our study, we have harnessed the power of deep learning, a prominent technique widely employed in natural language-related tasks, to tackle the challenging problem of toxin prediction. Leveraging the versatility of deep learning and large language architectures, we explored several existing models designed for sequence classification. Ultimately, we found that a custom-built artificial neural network incorporating LSTM layers, implemented using the Keras API, delivered the most promising performance, which is about the same performance as ESM2-t33. To mitigate overfitting concerns, we applied L2 regularization to ensure the generalizability and robustness of the model. In this case, we achieved a maximum AUC score of 0.93. Notably, the performance of the machine learning model was superior to deep learning. We employed machine learning methods on embeddings produced by ESM2-t33, ESM2-t6, ProtBERT, ProtBERT with Bi-LSTM, and BioBERT models. The embedding from ProtBERT with Bi-LSTM yielded the highest performance with an AUC of 0.957. Finally, our hybrid models combine the spectrum of both alignment-free and alignment-based methods. Notably, the MERCI-based hybrids, Extra Tree + MERCI-based ML hybrid model, and ANN-LSTM (fixed sequence length) + MERCI-based DL hybrid model have demonstrated superior performance and computational efficiency, exhibiting fast runtime processing. With AUC scores of 0.98 and 0.97, respectively, they are deployed on a web server for service to the community.

We established a comprehensive platform for protein and peptide classification as toxic or non-toxic. Our approach includes a user-friendly web server and an offline package for ToxinPred3.0. The web server integrates our accurate model, while the standalone package offers offline flexibility. Although our model shows promise, further enhancements are warranted. Continuous research will refine accuracy. We encourage researchers to adopt our method for designing improved protein and peptide therapeutics across diseases. This platform advances the field, aiding informed toxicological decisions and molecular design.

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Data availability statement

The datasets generated for this study can be accessed on the 'ToxinPred3.0′ web server at https://webs.iiitd.edu.in/raghava/toxinpred3/download.php. The source code for this study is publicly available on GitHub and can be found at https://github.com/raghavagps/toxinpred3.

CRediT authorship contribution statement

Anand Singh Rathore: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation. **Shubham Choudhury:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Akanksha Arora:** Writing – review & editing, Writing – original draft, Investigation, Data curation. **Purva Tijare:** Writing – review & editing, Software, Investigation. **Gajendra P. S. Raghava:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Project administration, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare no competing financial or non-financial interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2024.108926.

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